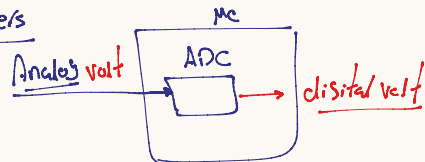
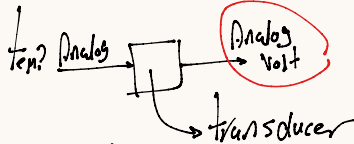


Transducers



* Intersect :: temp? / lighting / speed



* sensor → its transducer from any power to electricity.

* temp-sensing → object change structure Based on temp?

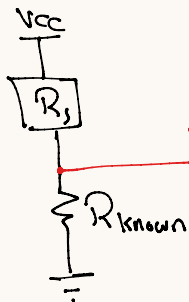
→ take this & create Resistor (Thermal Resistor)

Thermistor → R

* Apply temp

* Resistor change for value

→ Make voltage divide to change from Resistance into volt.



$$V_{out} = \frac{V_{cc} * R_{known}}{R_{known} + R_s}$$

* Thermistor → change value Based on temp?

→ Relation between Resistance & temp?

① NTC

negative temp Coefficient

$$R_s \propto \frac{1}{T}$$

* Available → low Cost

* Normal temp → (-55: 150)

② PTC

Positive temp Coefficient

$$R_s \propto T$$

* extreme Range

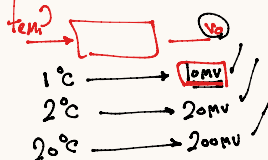
< -60	> 150
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* lm35 → sensor → temp → volt.

→ Normal Range ↓ low Cost | NTC

→ Have voltage divider

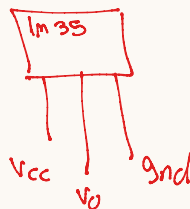
→ single ended →



1°C → 10mV

2°C → 20mV

20°C → 200mV



* How create Interface with lm35

- ① Get Digital volt from ADC
- ② by Calc (SW) → Convert from Digital volt to analog
- ③ Mapping between analog volt & temp

① Get Digital volt from ADC channel

uint16_t Digital volt = ADC_GetResult (0);

② Convert from Digital to Analog SV

uint16_t analog volt = Digital volt * step?

$$= (\text{Digital volt} * V_{ref}) / 1024$$

$$= (\text{Digital volt} * V_{ref}(mV)) / 1024$$

$$= ((\text{uint32}_t) \text{Digital volt} * V_{ref}(mV)) / 1024$$

$$\text{uint16}_t \text{ Analog volt} = ((\text{uint32}_t) \text{Digital volt} * V_{ref}(mV)) / 1024;$$

③ Mapping between temp & analog volt

→ temp = $\frac{\text{Analog volt}(mV)}{10}$

1°C → 10mV

Analog volt → temp

light $\rightarrow$ [R] $\rightarrow$ Resistance

$$\rightarrow R_L \propto \frac{1}{L}$$
$$* R_L \propto \frac{1}{L}$$
$$* V_0 \propto \frac{1}{R_L}$$

* $V_0 \propto L \rightarrow$ light intensity $\uparrow$ increase.

* Resistance decrease.

* You ↑ increase.

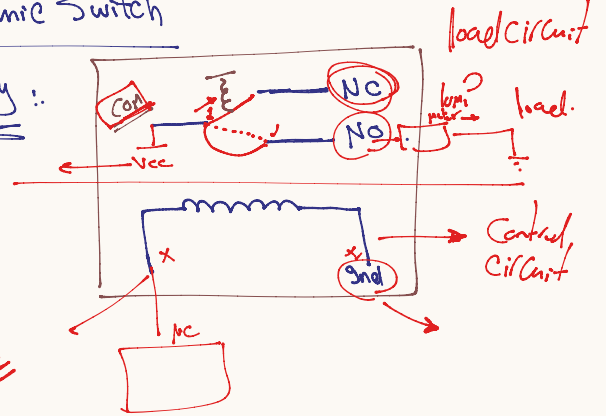
light intensity unit $\rightarrow$ lux

↓
Resistance

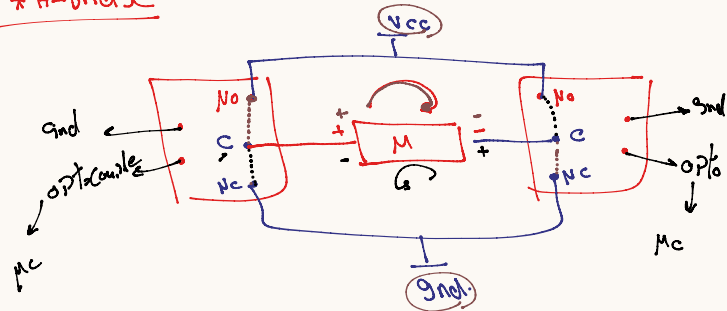
Based on voltage divider.

* How Inter face with MC $\rightarrow$ task ^{divider}

Relay :



* H-bridge



* tasks Drivers.

Tasks / Drivers:

① ADC driver (Interrupt Mode)

② IM35 Driver ③ IDR Driver

* Application

Migration

↳ Enter Password. ↙ Key Pad
↘ lcol.

sensor temp | IOR.

Sensor on? →
turn on motor

temp Display on LCD

sensor

→ state Display on LCD.

Morning
evening